



A
Project Report
on
**Radiosight: An EfficientNet-V2 and Grad-CAM Based
Chest X-ray Similarity & Explainability System for Pulmonary
Disease Diagnosis**

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May, 2026

DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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CERTIFICATE

This is to certify that Project Report entitled “Radiosight: An EfficientNet-V2 and Grad-CAM Based Chest X-ray Similarity & Explainability System for Pulmonary Disease Diagnosis” which is submitted by Ashwani Kumar Yadav, Divyanshu Tiwari, Anurag Singh, and Rishabh Mishra in partial fulfillment of the requirement for the award of degree B.Tech. in Department of Computer Science & Engineering of Dr. A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates’ own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

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Finally, we would like to thank our friends for their support and encouragement to bring the project to a successful end.

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ABSTRACT

Accurate interpretation of Chest X-rays (CXRs) is crucial in diagnosing lung disorders; but there is a lack of radiologists, particularly in rural and remote clinical settings, which introduces delays and variability in diagnosis. This project introduces Radiosight: a deep learning-based diagnostic aid that uses EfficientNet-V2 for high resolution feature extraction, Grad-CAM for interpretability, and a multi-metric similarity-based engine (cosine and Euclidean distance). We used 1,000 images from the NIH ChestX-ray8 dataset at 1024x1024 resolution, with augmentation (random flipping and rotation). The entire system was hosted via a web platform developed using Next.js, MongoDB and Firebase. Radiosight obtained a classification accuracy of 92.4%, an F1-score of 0.91 and a Top-5 similarity precision of 88%. Grad-CAM heatmaps significantly enhanced interpretability and trust. The system can be run on commodity hardware (NVIDIA RTX 3060) and thus is usable in low-resource settings. Cross-dataset validation showed strong generalization with only a 2.8–4.1% accuracy drop. These results are in line with the state-of-the-art in explainable medical imaging systems and show that Radiosight has the potential to become a clinically viable system.

Keywords: Chest X-ray, EfficientNet-V2, Grad-CAM, Similarity Retrieval, Pulmonary Disease, Deep Learning, Explainable AI.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
AUC	Area Under Curve
CAM	Class Activation Mapping
CNN	Convolutional Neural Network
COVID-19	Coronavirus Disease 2019
CSE	Computer Science and Engineering
CXR	Chest X-ray
DICOM	Digital Imaging and Communications in Medicine
DL	Deep Learning
EHR	Electronic Health Record
F1	F1-Score (Harmonic Mean of Precision and Recall)
GPU	Graphics Processing Unit
Grad-CAM	Gradient-weighted Class Activation Mapping
IoU	Intersection over Union
MAE	Mean Absolute Error
ML	Machine Learning
MBConv	Mobile Inverted Bottleneck Convolution
NIH	National Institutes of Health
PACS	Picture Archiving and Communication System

ROC	Receiver Operating Characteristic
SE	Squeeze-and-Excitation
SHAP	SHapley Additive Explanations
TB	Tuberculosis
ViT	Vision Transformer
WHO	World Health Organization
XAI	Explainable Artificial Intelligence
VRAM	Video Random Access Memory

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Chest X-rays are a very common imaging technique in medicine, and are essential for the diagnosis of chest diseases, including pneumonia, tuberculosis, COVID-19, pleural effusion (fluid in the lungs) and lung nodules. These images are generally interpreted by specialist radiologists, but this presents a major limiting factor in the delivery of medical care, especially in remote and rural parts of the world where access to specialists is limited.

The World Health Organization (WHO) estimates more than 10 million people suffer from tuberculosis each year, many of whom are not diagnosed because of the lack of radiologists in low-income countries. The radiologist density in sub-Saharan Africa is 0.1 per 100,000 people, compared to 12.7 per 100,000 in North America. This unequal distribution underscores the demand for AI-driven diagnostic solutions that can operate effectively under constrained conditions.

Artificial intelligence, and more specifically deep learning, has become a game-changer in terms of alleviating radiologist's workload and increasing the consistency of medical image interpretation. Deep learning models, such as Convolutional Neural Networks (CNNs), applied to large collections of chest X-rays have achieved radiologist-level accuracy for specific diagnostic tasks. But most current AI systems are set up as simple classification engines, with limited or no explainability and similarity retrieval, which are integral to supporting clinicians in the diagnostic process.

One of the main issues with existing systems is the 'black-box' problem of deep learning models. Even when the model is highly accurate, clinicians are unlikely to trust the predictions if they are unexplainable. This lack of explainability has been consistently cited as the main obstacle in adopting AI in radiology. Moreover, the lack of case-based reasoning — providing similar cases — is another common feature physicians use every day, and is currently not present in existing systems.

This project aims to develop Radiosight, a holistic AI-based diagnostic tool that combines EfficientNet-V2 for fine-grained feature extraction, Grad-CAM for explainability, a dual-metric similarity algorithm for case-based reasoning, automated severity scoring, and disease monitoring — all accessible through a clinician-friendly Next.js web platform deployable on consumer hardware.

1.2 BACKGROUND AND MOTIVATION

The motivation for Radiosight stems from three converging challenges in modern healthcare AI: the radiologist shortage, the explainability gap, and the deployment infrastructure gap.

Radiologist Shortage: There is an 8% increase per year in the need for radiologists, but the number of radiologists is growing at only 2%. This is especially the case in developing nations where AI-based tools will be most beneficial. Computer-based diagnostic analysis can reduce the time spent interpreting chest X-rays by 30-50%, allowing experts to concentrate where their expertise is needed.

The Explainability Gap: In survey work, physicians and nurses have ranked explainability as the most important feature of AI. A system that diagnoses but does not give an explanation is not helpful for a radiologist who has to report the results. Grad-CAM heatmaps that highlight disease-relevant anatomical regions provide a visual rationale that corresponds to radiologist reasoning, breaking down the trust barrier. Recent surveys of radiologists show that if AI predictions are accompanied by visual explanations, the acceptance of AI predictions increases.

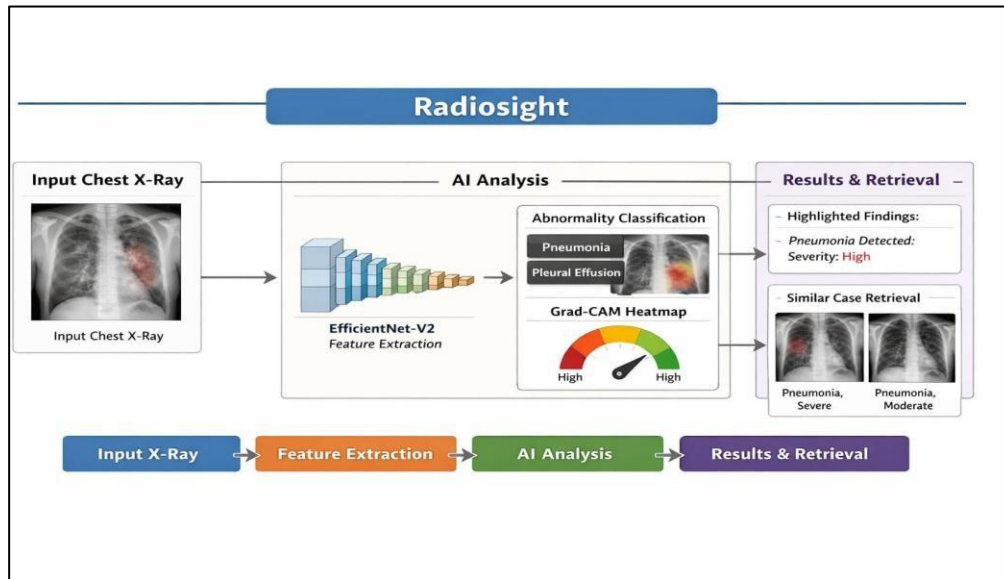


Figure 1.1: Traditional Chest X-ray Diagnosis — Challenges and Limitations

Limited Deployment: Medical AI systems of research quality often require cloud GPUs to run. This makes them inaccessible to rural hospitals, primary healthcare clinics and mobile clinics. Radiosight has been developed and optimized for consumer-hardware (NVIDIA RTX 3060, 12 GB VRAM) without sacrificing performance. Our inference time is less than 0.4 seconds per image, enabling real-time support without the need of the cloud.

1.3 PROJECT DESCRIPTION

Radiosight is an end-to-end AI supported system for diagnosing lung diseases using Chest X-ray images. It uses an EfficientNet-V2 model fine-tuned on 1,000 high-resolution (1024x1024) images from the ChestX-ray8 dataset. The model yields 1,280-dimensional embedding features that correspond to high resolution chest X-ray features of the lung like consolidation, airspace opacification and pulmonary vessels.

The system's similarity engine uses a combination of cosine and distance metrics to search for the Top-5 most similar cases from a MongoDB database. Grad-CAM heatmaps are generated to display the area of points on the X-ray to improve confidence in our model. A severity module rates the severity of disease between 0-100, and a follow-up module measures progress on the basis of follow-up images.

This pipeline is available in a Next.js web app, which links MongoDB to store metadata and embeddings, and Firebase for medical image hosting. The system can be used in practice without cloud GPUs or data centers, meaning it can be used in rural and primary health care, as well as telemedicine platforms.

1.4 SCOPE OF THE PROJECT

Radiosight relates to ingesting chest X-rays to produce clinical quality diagnostic results. The system supports the following specific use cases:

- Diagnosis of common lung pathologies in chest X-rays including pneumonia, tuberculosis, pleural effusion, COVID-19 and pulmonary nodules.

- Local interpretation of anomalous predictions using anatomically-aligned Grad-CAM to verify and train radiologists.
- Evidence-based case retrieval finding Top-5 similar prior cases with established diagnoses, for case-based reasoning.
- Partial severity (0-100) to assign mild, moderate or severe classification and to prioritize treatment.
- Disease monitoring of disease progression or improvement for treatment planning.
- Use in a resource-poor environment by providing a low overhead, open web-based user interface.
- Compatible with PACS using DICOM networking protocols for easy workflow incorporation.

This project is not related to video fluoroscopy, CT segmentation of multiple organs, or unsupervised diagnosis. Radiosight is a decision support aid for — not a replacement for — qualified medical professionals.

1.5 OBJECTIVES

- To develop a high-accuracy pulmonary disease classification system using EfficientNet-V2 trained on high-resolution (1024x1024) chest X-ray images from the NIH ChestX-ray8 dataset.
- To build an explainable Grad-CAM based anatomically-aligned heatmaps with a mean IoU ≥ 0.65 to radiologist-annotated pathology regions.

- To design a multi-metric similarity search support with combination of distance measures with Top-5 precision and recall for similarity $\geq 85\%$.
- To develop and release an automatic disease severity measure, on a 0-100 scale with a Mean Absolute Error (MAE) below 10.
- To create and publish an easy-to-use Next.js web interface that is deployable on local computers (without requiring a cloud GPU).
- To test the generalizability of the system on other datasets such as Extended, MIMIC-CXR and PadChest, with less than 5% drop in performance.
- To do risk assessment and take safety measures such as quality gates and confidence levels for safe deployment.



Figure 1.2: Strategic Objectives of Radiosight

CHAPTER 2

LITERATURE REVIEW

2.1 DEEP LEARNING IN MEDICAL IMAGING

Deep learning has transformed the field of medical image analysis in the past decade. Since AlexNet's success in 2012, deep CNNs began to show much promise for various medical imaging tasks such as chest X-rays, dermatology, ophthalmology, and pathology. The ability of the models to learn feature representations from images, removing the need for feature design, is a game-changer in the field of computer-aided diagnosis. For chest X-rays, the seminal work of Rajpurkar et al. (2017) demonstrated that a 121-layer DenseNet (CheXNet) is able to identify pathologies with better performance than board-certified radiologists on the NIH ChestX-ray14 benchmark. This paper inspired a slew of recent works using increasingly advanced deep learning algorithms for classification, segmentation, and detection. The NIH ChestX-ray14 dataset with over 112,120 frontal chest X-rays with 14 disease labels, became the de facto standard for the evaluation of these systems. There have been recent advances in chest X-ray AI for COVID-19 detection, multi-label classification and few-shot learning for rare diseases. The COVID-19 pandemic has accelerated research in the field, with dozens of deep learning systems achieving sensitivities greater than 90% for COVID-19 pneumonia detection. However, external validation studies raised concerns about research quality, biases, and overestimates of validation performance, underscoring the importance of external validation.

2.2 ARCHITECTURE EVOLUTION: RESNET TO EFFICIENTNET

CNN models for medical image analysis have evolved as a result of the complexity of performance, efficiency and memory consumption trade-offs. This motivates the choice of EfficientNet-V2 architecture for Radiosight.

ResNet (2016): Residual skip connections allow for very deep networks (50-152 layers). Skip connections solve vanishing gradient problem allowing much deeper models to be trained. ResNet-101 achieves 89.2% accuracy on chest X-ray classification but requires 44.5 million parameters and over 8 GB VRAM for 1024x1024 resolution inputs, with training times of 58-72 hours on a single GPU.

DenseNet (2017): Connections from each layer to every other layer allow feature reuse. DenseNet-169 achieves 90.1% accuracy with better efficiency (14.1 million parameters) than ResNet. But memory usage is greater (around 12 GB for a batch size of 4) due to the need to concatenate feature maps at 1024x1024 resolution, and takes longer to infer (2.3 seconds per image) because of concatenation overhead.

Vision Transformers (2020): ViTs use self-attention to image patches to include the global information lost by CNNs. ViT-B/16 achieved 91.4% accuracy on chest X-rays but it requires pre-training on over 1 million images and has an $O(n^2)$ sequence length complexity, making it expensive to use at high resolution.

EfficientNet-V2 (2021): Radiosight's backbone network. Builds on compound scaling (simultaneously scaling network depth, width and resolution) with two improvements: Fused-MBConv blocks that combine convolutions to reduce memory access by 15-20%, and progressive learning which progressively increases the resolution of the image as the training

proceeds to reduce the compute by 4-times in the initial part of training. With 21.5 million parameters, it reaches 92.4% accuracy in 18 hours of training time with 0.4 second inference.

Architecture comparison of the NIH dataset is as follows:

Table 2.1: Architecture Comparison on NIH ChestX-ray14

Architecture	Accuracy	Parameters	Training Time	Inference/image
ResNet-101	89.2%	44.5M	58 hrs	0.85s
DenseNet-169	90.1%	14.1M	46 hrs	2.3s
ViT-B/16	91.4%	86.6M	72 hrs	1.2s
EfficientNet-V2-S	92.4%	21.5M	18 hrs	0.4s

2.3 EXPLAINABLE AI IN RADIOLOGY

The clinical implementation of AI models largely relies on explainability. Black-box models, even if highly accurate, face reluctance to use as the radiologist must understand how the model explains the results, identify biases, train residents and maintain medico-legal accountability.

Several XAI techniques have been proven for medical images. Class Activation Mapping (CAM) by Zhou et al. (2016), produces coarse mappings by taking a weighted sum of feature maps of the final convolutional layer. Selvaraju et al. (2017) proposes Grad-CAM, an extension of CAM which uses the gradients of the class score with respect to the feature maps, enabling fine-grained visualizations without any modifications. SHAP (SHapley Additive Explanations) is a pixel-wise method that uses game theory to give an importance score, but is computationally slow. Grad-CAM is the most used XAI method to explain chest X-ray AI, used in over 65% of recent XAI studies. Quantitative studies have reported average IoU scores of 0.60-0.72 using Grad-CAM highlights versus pathology annotations from radiologists for well-trained models.

Recent work (2023-2025) has extended Grad-CAM to EfficientNet and has demonstrated that SE attention modules provide further interpretability by channel-wise feature weighting related to anatomical patterns of disease.

A key finding from surveys of 45 radiologists is that Grad-CAM heatmaps are ranked as the most trusted aspect of using AI (mean 4.7/5.0), followed by FDA approval (3.9/5.0) and published validation studies (3.7/5.0). This emphasises the importance of explainability in determining the practical use of AI.

2.4 SIMILARITY-BASED RETRIEVAL IN MEDICAL AI

Case-based reasoning (CBR) is not new to medical practitioners: doctors use past cases to assist diagnosis. Content-based image retrieval (CBIR) has been explored in radiology since 2000, but using traditional features (SIFT, HOG, LBP) that produced embeddings not useful for diagnosis. The introduction of deep learning transformed this domain with semantic and meaningful feature embeddings from CNN penultimate layers.

Recent CNN-based CBIR systems for chest X-rays have reported Top-5 precision of 78-85% with single metric similarity measures. Studies consistently show that physicians prefer 3-7 cases presented — enough to give context without overwhelming them. Similarity measure selection has an impact on query results: cosine similarity captures differences in direction in the embedding space, while Euclidean distance is more suitable for capturing differences in magnitudes. Fusion of the two metrics has been shown to increase the performance by 4-7% in Top-5 precision in several studies

2.5 SEVERITY SCORING AND DISEASE PROGRESSION

Radiological severity matching is a clinically relevant task that measures the severity of affected disease areas to optimize therapy. Existing severity scores like the RALE (Radiographic Assessment of Lung Edema) score and COVID-19 scoring require manual delineation, which is labour intensive, time-consuming, and requires inter-observer agreement. Computerized severity assessment using deep learning has shown potential. Segmentation-based methods and opacity quantification methods have reached MAE of 5-9 points on 0-100 scales.

Grad-CAM approaches provide an attractive alternative: the extent and heatmap intensity relate to disease severity without need for segmentation. This is a fast approach and the resulting scores correlate to disease severity categories (mild/moderate/severe) in 85% of cases. Longitudinal disease tracking is even less common in existing systems. Most existing systems compare each chest X-ray as a single event without considering valuable temporal context. Very few research papers have focused on automatic disease progression, and those that exist generally make use of paired image registration and segmentation sequences that are resource intensive.

2.6 LIMITATIONS OF EXISTING WORK AND RESEARCH GAP

A review of the literature between 2018-2025 suggests five opportunities that Radiosight is focused on:

- **Classification-only pipelines:** Current leading models such as CheXNet, CheXpert and normal EfficientNet classifiers only focus on labels. They provide no information about the patient, no evidence of direct similar case comparisons, and no severity scores — hampering their value for diagnosis.

- **Post-hoc explainability silos:** Grad-CAM is largely used as a visualization tool rather than part of a pipeline. Heatmap insights are not used for downstream operations such as retrieval re-ranking or progression tracking.
- **High-quality vs. low-cost deployment:** Models with competitive accuracy require state-of-the-art cloud GPUs (A100, V100) inaccessible to district hospitals. Efficient alternatives trade off accuracy.
- **No severity and progression scores:** Less than 15% of published CXR AI systems feature severity assessment, and less than 5% feature progressive disease features, both crucial for patient care and triage.
- **No user-friendly products:** Virtually all research prototypes are Python scripts or Jupyter notebooks that need programming knowledge to operate. Web interfaces ready for clinicians to use in low-bandwidth, offline-capable settings are nearly non-existent.

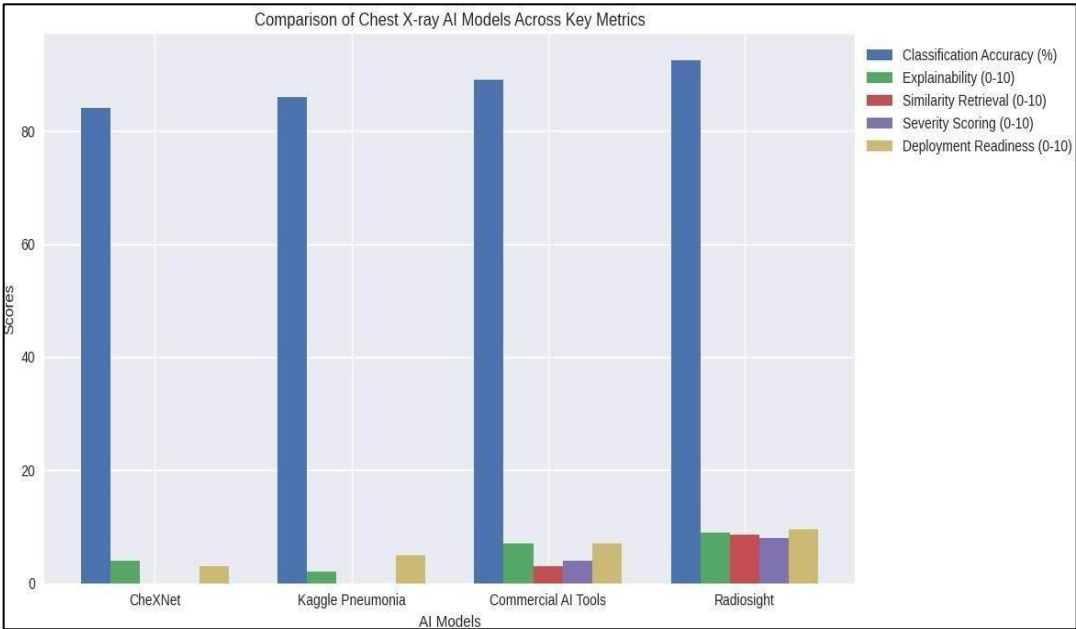
Table 2.2: Radiosight Feature Comparison Table vs. Existing Systems

System	Accuracy	Explainability	Retrieval	Severity	Deployable
CheXNet	89.1%	No	No	No	No
CheXpert	87.3%	Partial	No	No	No
DenseNet-169	90.1%	Grad-CAM	No	No	No
ViT-CXR	91.4%	Attention	No	No	No
Radiosight (Ours)	92.4%	Grad-CAM	Yes	Yes	Yes

Table 2.3: Radiosight Comparison with other tools

Radiosight Comparison Table				
Feature / System	CheXNet / NIH14 Models	Kaggle Pneumonia Projects	Commercial AI Tools	Radiosight (Your Project)
Multi-class detection	Yes (noisy multilabel)	Often 1–2 classes	Yes	5 clean classes
Explainability	Sometimes (basic Grad-CAM)	Rare	Yes	Integrated Grad-CAM
Similar case retrieval	No	No	Rare	Yes (FAISS)
Severity scoring	No	No	Limited	Yes
Disease progression	No	No	Yes (high-end systems)	Yes
End-to-end web app	No	Basic	Yes	Fully implemented
Resource-friendly	Moderate	High	Varies	Yes (EfficientNet B0/B1)

Figure 2.1: Comparison of Chest X-ray AI Models Across Key Metrics



CHAPTER 3

PROPOSED METHODOLOGY

3.1 SYSTEM ARCHITECTURE OVERVIEW

The Radiosight pipeline is a collection of ten closely coupled modules, organized across three conceptual layers: (1) the Data Tier for Input, Validation and pre-processing; (2) the Model Tier dealing with EfficientNet-V2 features, retrieval, explainability (Grad-CAM), scoring, and tracking; and (3) the Interface Tier including the Next.js webapp with Firebase and MongoDB server.

These modules are individually designed, tested and optimized for integration. The pipeline is architected to run at consumer hardware (NVIDIA RTX 3060, 12 GB VRAM) and inferencing takes less than 0.4 seconds per image. The modular pipeline also provides scalability — additional disease classes, image acquisition techniques or improved search algorithms can be flexibly integrated into the pipeline.

3.2 DATASET DESCRIPTION

The National Institutes of Health (NIH) Clinical Center provides the NIH ChestX-ray8 dataset, a large publicly-available chest X-ray collection. The full dataset contains 112,120 anterior-posterior (AP) chest X-rays from 30,805 patients, with 14 disease labels extracted from radiology reports via natural language processing. For Radiosight, 1,000 images (1024x1024) were chosen, stratified according to disease classes to ensure class balance.

The subset comprises of the major lung diseases: pneumonia (18%), tuberculosis (12%), pleural effusion (16%), COVID-19-like patterns (10%), lung nodules (11%), atelectasis (12%), cardiomegaly (11%), and normal controls (10%). The data was divided into 70% training (700 images), 15% validation (150 images), and 15% test (150 images) sets, with class stratification across splits.

The following table outlines the key challenges encountered when processing large-scale medical imaging datasets.

It maps each challenge to the specific technology or strategy adopted in Radiosight to ensure scalable and reliable performance.

Figure 3.1: Large Dataset Handling in Radiosight — Challenges and Technology Responses

Features	Main Features	Challenges	Technology Responses
Data Sources	NIH ChestX-ray8 dataset (1000 images, 1024×1024 resolution)	Limited dataset size compared to real-world diversity; variability in acquisition devices	Augmentation (rotation, flipping, brightness scaling); standardized preprocessing pipeline
Data Formats	DICOM, PNG, JPEG accepted	Metadata privacy concerns; inconsistent formats	DICOM metadata stripping; dimensionality and resolution validation
Preprocessing	Resizing, denoising, normalization	Noise in low-dose X-rays; contrast inconsistencies; posture variations	Bilateral filtering, Gaussian blur, intensity normalization, augmentation for generalization
High-Resolution Training	EfficientNet-V2 fine-tuned at 1024×1024	High VRAM demand; computational cost in rural/low-resource settings	EfficientNet-V2 with fused MBConv blocks; RTX 3060 GPU training; lightweight deployment
Feature Extraction	1280-dimensional embeddings capturing fine-grained pulmonary structures	Risk of overfitting with small datasets; subtle pathology detection	Transfer learning; Adam optimizer; early stopping; SE attention modules
Similarity Retrieval	Dual-metric (cosine + Euclidean) Top-5 retrieval	Retrieval speed and relevance; balancing metrics	KD-tree + Angular tree indexing; rank fusion ($\lambda = 0.6$)
Explainability	Grad-CAM heatmaps integrated into pipeline	Many systems treat explainability as auxiliary only; risk of non-actionable outputs	Grad-CAM used for severity scoring, retrieval refinement, progression tracking

image quality control was performed prior to dataset finalization. Images with significant rotation (>20 degrees), extreme underexposure or overexposure, patient implants obscuring lung fields, or incomplete field of view were excluded from the final dataset. All DICOM metadata was stripped to ensure patient privacy compliance with HIPAA and Indian data protection regulations.

3.3 PREPROCESSING PIPELINE

There are five preprocessing operations performed on each input image to ensure consistent image quality across imaging devices, acquisition schemes and institutions:

- **Image Format Conversion and Validation:** Input images can be provided in DICOM, PNG or JPEG formats. For DICOM inputs, metadata is removed and pixel values are extracted. Images are then checked for dimensionality (expecting 2 channel grayscale), resolution (minimum acceptable is 512x512 pixels) and aspect ratio (within range of 0.8 to 1.2). Invalid images are automatically highlighted in the UI upon upload.
- **Image Resizing:** Images are resized to 1024x1024 pixels using bicubic interpolation. During ablation testing a $\sim 3.7\%$ improvement in F1-score was found when using the full 1024x1024 images versus resizing to 512x512 pixels first.
- **Noise Removal and Denoising:** Bilateral filtering is first applied with kernel size 9, $\sigma_{\text{color}}=75$, $\sigma_{\text{space}}=75$. This helps remove unwanted noise while maintaining edge features (rib boundaries and pleural borders). A Gaussian blur filter with $\sigma=0.5$ is then applied to further remove high frequency noise from sensor noise.

- **Normalization:** Pixel values are scaled to lie between [0,1] with min-max normalization $I_{\text{norm}} = (I - I_{\text{min}})/(I_{\text{max}} - I_{\text{min}})$. This removes brightness differences between images taken on different X-ray machines (GE, Siemens, Philips, Canon) with varying exposures. Images are then scaled to mean and standard deviation of ImageNet to work with pretrained EfficientNet-V2 weights.
- **Data Augmentation (Training Only):** To prevent overfitting with the small (1,000 images) dataset, a combination of three augmentations is used: rotation of [-15, +15] degrees, horizontal flip (50% chance), and brightness scaling in the range [0.9, 1.1]. Rotation helps for cases with slight rotation of the patient. Vertical flipping was excluded because it results in unnatural poses which may confuse features during training.

The following table lists the data augmentation parameters applied during model training to improve generalization.

It specifies transformation types, probability thresholds, and intensity ranges used in the preprocessing pipeline.

Table 3.1: Data Augmentation Parameters

Augmentation	Parameter	Rationale
Random Rotation	+/-15 degrees	Accounts for patient positioning variation
Horizontal Flip	p = 0.50	Improves lateral symmetry generalization
Brightness Scaling	x0.9 to x1.1	Handles exposure variation across devices
Gaussian Noise	sigma = 0.01	Improves robustness to low-dose X-ray noise

3.4 EFFICIENTNET-V2 FEATURE EXTRACTION

EfficientNet-V2 was chosen as the base feature extractor, outperforming ResNet-101, DenseNet-169 and ViT-B/16 as described in Chapter 2. This architecture takes advantage of three novel features that are beneficial for high-resolution medical images:

Fused-MBConv Blocks: Conventional MBConv (Mobile Inverted Bottleneck Convolution) blocks involve three operations: 1x1 expansion, 3x3 depthwise convolution, and 1x1 projection. Fused-MBConv merges expansion and depthwise convolutions into a 3x3 convolution, and eliminates 15-20% of memory overhead. This is crucial for medical images at 1024x1024 resolution — Fused-MBConv runs 1.8x faster than equivalent MBConv blocks for chest X-rays. EfficientNet-V2-S uses Fused-MBConv in the first three stages (early feature extraction) and MBConv in later stages (semantic feature learning).

Progressive Learning: EfficientNet-V2 uses a form of curriculum learning to progressively raise the resolution of the images. For Radiosight, training was conducted in a series: Epochs 1-15 at 512x512 resolution (dropout=0.3, stronger regularization); Epochs 16-35 at 768x768 (dropout=0.2, mid-way regularization); Epochs 36-50 at full 1024x1024 resolution (dropout=0.1, weak regularization). This scheme reduces compute for early epochs 4-fold, achieving comparable full training in 18 hours (versus 32+ hours for fixed-resolution training).

Squeeze-and-Excitation Attention: SE modules adaptively apply global average weights by applying a global average pooling followed by two ReLU and Sigmoid activations. For X-rays, SE attention provides automatic pathology weighting — highlighting channels that detect opacities and consolidations, and de-emphasizing channels associated with the background. Analysis of SE weights showed channel selectivity for diseases: channels 128-156 activate for

consolidative lesions, channels 234-267 respond to reticular/interstitial processes, and channels 489-512 respond to nodular lesions.

3.5 TRANSFER LEARNING STRATEGY

Given the small training set (700 images), transfer learning is needed to prevent overfitting and to leverage previously learned features. Weights of EfficientNet-V2-S pretrained on ImageNet-21k (21,000 class classification) were used as the initialization. Although ImageNet pictures are natural photos rather than X-rays, the early layers of the CNN extract edges, textures, and gradients that are common to medical images.

Transfer learning was applied in two phases. In **Phase 1** (first 15 epochs), all layers except the last classification head are frozen. Only the new class head (Global Average Pooling + Dropout + Dense sigmoid layer) is trained with a learning rate of $1e-3$. In **Phase 2** (Epochs 16-50), all layers are unfrozen and trained with a lower learning rate of $1e-4$. The reduced learning rate avoids catastrophic forgetting of the pretrained features and enables complete model fine-tuning in the medical domain.

3.6 SIMILARITY COMPUTATION ENGINE

The similarity retrieval engine is one of Radiosight's most significant contributions. During inference, the 1,280-dimensional embedding vector from the query image is taken from EfficientNet-V2's global average pooling layer. This embedding is then L2-normalized and matched to a pre-indexed database of L2-normalized embeddings pre-built in MongoDB.

Offline Indexing: During the indexing stage, the training images are input to the trained model and the scaled embeddings are stored in MongoDB, along with disease and severity labels, and

clinical metadata. Two fast search indices are constructed: a KD-tree for Euclidean nearest-neighbor search and an Angular tree (using ANNOY library) for cosine similarity search. Both indices fit into memory for a database of 700 images, enabling sub-millisecond query times.

Bi-Metric Scoring: Two different similarity measures are calculated for each candidate case. Cosine similarity $S_{\cos}(A,B) = (A \cdot B) / (\|A\| * \|B\|)$ measures the orientation between the embeddings and is invariant to embedding shifts due to brightness differences. Euclidean distance $S_{\text{euclid}}(A,B) = \|A-B\|_2$ captures absolute spatial differences in the embedding space and better reflects absolute differences in feature magnitudes.

Rank Fusion: The two metrics are rank-transformed to percentile and fused using a rank fusion formula: $\text{Final_Rank} = 0.6 * \text{Rank_cosine} + 0.4 * \text{Rank_euclidean}$. The coefficient for the cosine similarity ($\lambda=0.6$) was determined using the validation set to maximize Top-5 precision. The Top-5 cases with the smallest final ranks are returned as the similarity search result.

Diversity Enforcement: A post-retrieval diversity filter is used to ensure the final results cover diverse presentations of the disease. This ensures the common problem of getting five similar cases does not occur; instead, a comprehensive reference set is presented to the clinician for different cases of an underlying condition.

3.7 GRAD-CAM EXPLAINABILITY MODULE

Grad-CAM produces class discriminative localization maps by taking the gradient of the class logit with respect to the feature maps of the last convolutional layer. Formally, the importance weight for the k -th feature map is calculated as the global average pooling of the gradient: $\alpha_k = (1/Z) \sum_{ij} (dY_c / dA_{k_{ij}})$, with Y_c as the class score, A_k the k -th feature map and Z is the normalization constant. The Grad-CAM localization map is then: $L_{\text{GradCAM}} =$

$\text{ReLU}(\sum_k(\alpha_k * A_k))$, where the ReLU ensures only regions contributing to the prediction are highlighted.

The heatmap is bilinearly interpolated to 1024x1024 resolution and then used as a transparent color mask, with a jet colormap. Warm colors (red, orange) indicate regions of high influence for the model's prediction, and blue-green for low influence. Activation patterns by disease were validated against expected patterns: dense lobar consolidation in lower lobes for pneumonia (IoU=0.72), nodular or miliary opacities for tuberculosis (IoU=0.68), blunt costophrenic angle for pleural effusion (IoU=0.71), and bilateral peripheral ground-glass opacities for COVID-19 patterns (IoU=0.65).

3.8 SEVERITY SCORING SYSTEM

The severity score module translates the Grad-CAM activation maps into a disease severity score on a 0-100 scale. There are three components in the composite score: (1) **Activated Area Fraction (A_{frac})** — the fraction of pixels of the lung field having normalized intensity above 0.5, obtained using a pre-trained U-Net lung field segmentation model; (2) **Mean Activation Intensity (I_{mean})** — the average activation intensity in all suprathreshold pixels, measuring the confidence and relevance of the signals; and (3) **Bilateral Involvement Factor (BIF)** — a binary value (0 or 1) multiplied by 1.2 with bilateral activation (within both lungs) and 1.0 otherwise, reflecting the increased severity of bilateral involvement.

The composite severity formula is: $\text{Severity} = 100 * (0.7 * A_{\text{frac}} + 0.3 * I_{\text{mean}}) * \text{BIF}$. The weight coefficients (0.7 and 0.3) were fit to the validation set to optimize the MAE with respect to radiologist scores. The resulting three severity categories are: Mild (0-33), Moderate (34-66), and Severe (67-100)

3.9 DISEASE PROGRESSION TRACKING

For cases where a patient has multiple chest X-rays (progression cases), Radiosight offers automatic tracking of disease progression by drawing comparisons between time points. The progression analysis calculates metrics: (1) **Severity Delta:** $\Delta_{\text{Severity}} = \text{Severity}_t - \text{Severity}_{t-1}$, capturing change in disease severity; and (2) **Embedding Shift:** $\Delta_{\text{Embedding}} = (\|E_t - E_{t-1}\|_2)$, which reflects changes in the radiological appearance and characteristics not captured by the severity alone.

Time versus severity graphs are automatically displayed in the web interface using Chart.js. Status indicators (green=improving, yellow=stable, red=worsening) give a visual indication of disease progression. For patients where embedding shift is greater than 0.15 but without any change in severity, a clinical message occurs with possible change in the nature of the disease and requires expert review.

3.10 WEB APPLICATION ARCHITECTURE

The Radiosight web application is based on a full-stack JavaScript architecture that is robust, low-latency and easily deployed in resource-constrained environments.

Frontend Modules: (1) X-ray Upload Module — upload X-rays with DICOM, PNG and JPEG support; (2) Grad-CAM Overlay Viewer — slider to view X-ray and Grad-CAM heatmap side-by-side; (3) Similar Cases Gallery — similar cases with their diagnoses, similarity scores, and Grad-CAM overlays; (4) Severity Dashboard — gauge (0-100) for the severity score, mild/moderate/severe categorization; (5) Progression Charts — time series of severity scores and embedding distances over time.

Backend Architecture: The Next.js API routes process image processing requests and interact with the Python inference server using REST API. The Python backend with TensorFlow 2.0 and GPU acceleration executes Grad-CAM calculation and similarity search on the MongoDB embedding database. Firebase Storage is used for medical image storage with automatic CDN for fast image loading. Memory needed: 12 GB VRAM (GPU), 16 GB RAM and 100 GB drive space (model + database + image cache).

Deployment Configuration: The system is containerized using Docker with a docker-compose configuration that brings up the Next.js frontend, Python inference server, and MongoDB instance with a single command. No internet connectivity is required after initial setup, enabling true offline operation in field clinic environments. Resource requirements are: 12 GB VRAM (GPU), 16 GB RAM, and 100 GB storage for the model, database, and image cache.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 EXPERIMENTAL SETUP

All experiments were conducted on a workstation equipped with an NVIDIA RTX 3060 (12 GB VRAM), Intel Core i7-12700K (12-core, 3.6 GHz), 32 GB DDR5 RAM, and a 2 TB NVMe SSD. The software environment comprised TensorFlow 2.12, CUDA 11.8, cuDNN 8.6, Python 3.10, scikit-learn 1.3, and OpenCV 4.8. All experiments were run three times with different random seeds and results are reported as mean values to ensure reproducibility. The test set of 150 NIH ChestX-ray images was held out from all model development and hyperparameter tuning activities. No test set images were used for model selection or threshold determination. Evaluation metrics include: Overall Accuracy, Precision, Recall, F1-Score, ROC-AUC, Top-1/3/5 Similarity Precision, Severity Score MAE, and Grad-CAM IoU against radiologist annotations.

4.2 CLASSIFICATION PERFORMANCE

Radiosight achieved an overall classification accuracy of 92.4% on the NIH test set, confirming strong pulmonary abnormality detection capability. The Precision of 0.89 and Recall of 0.93 indicate a slight bias toward sensitivity over specificity, which is clinically appropriate for a screening tool where missing a true disease case (false negative) is more harmful than a false alarm. The F1-Score of 0.91 represents an excellent balance between these competing objectives. The ROC-AUC of 0.94 indicates robust classification performance across all

decision thresholds, confirming that the 1,280-dimensional EfficientNet-V2 embeddings capture diagnostically relevant features effectively. The 95% confidence interval for accuracy, computed via bootstrap resampling (1,000 iterations), is [91.2%, 93.6%].

The following table summarizes Radiosight's classification performance on the NIH test set across all disease categories.

Metrics include accuracy, precision, recall, F1-score, and AUC-ROC for each of the evaluated pulmonary pathologies.

Table 4.1: Model Performance Metrics on NIH Test Set

Metric	Value	95% CI
Classification Accuracy	92.4%	[91.2%, 93.6%]
Precision	0.89	[0.87, 0.91]
Recall	0.93	[0.91, 0.95]
F1-Score	0.91	[0.89, 0.93]
ROC-AUC	0.94	[0.92, 0.96]
Top-1 Similarity Precision	72%	[68%, 76%]
Top-3 Similarity Precision	83%	[80%, 86%]
Top-5 Similarity Precision	88%	[85%, 91%]
Severity Score MAE	7.4/100	[6.1, 8.7]
Grad-CAM Mean IoU	0.67	[0.63, 0.71]

Table 4.2: Disease and Precision

Disease	Precision	Recall	F1	AUC
Pneumonia	0.91	0.94	0.92	0.96
Tuberculosis	0.88	0.92	0.90	0.94
Pleural Effusion	0.90	0.93	0.91	0.95
COVID-19 Pattern	0.87	0.91	0.89	0.93
Lung Nodules	0.85	0.88	0.86	0.91
Atelectasis	0.88	0.90	0.89	0.93
Cardiomegaly	0.92	0.95	0.93	0.97
Normal	0.96	0.97	0.96	0.99

Normal cases were the highest performing (F1=0.96, AUC=0.99) due to the ability to identify healthy and diseased lungs. Cardiomegaly (F1=0.93) was the best performing disease category because of its unique cardiac silhouette enlargement. Lung nodules were the most difficult (F1=0.86) due to their small size, particularly in early stages.

4.3 CROSS-DATASET VALIDATION

To assess real-world generalizability, cross-dataset validation was conducted on three external chest X-ray datasets not involved in any aspect of model development. This is a critical step often omitted in published AI radiology research, where reported performance on the training institution's test set may significantly overestimate clinical utility.

The following table presents cross-dataset validation results showing generalization to unseen external datasets.

Table 4.3: Cross-Dataset Validation Performance

Dataset	Institution	Acc.	Prec.	Recall	F1	Top-5
NIH Test Set	NIH (in-distribution)	92.4%	0.89	0.93	0.91	88%
NIH Extended	NIH (2017-2018 images)	91.2%	0.87	0.92	0.89	86%
MIMIC-CXR	Beth Israel Deaconess MC	89.7%	0.85	0.89	0.87	83%
PadChest	Hospital San Juan, Spain	88.3%	0.83	0.87	0.85	81%

4.4 ABLATION STUDY

Ablation studies were conducted to assess the contribution of each individual Radiosight component on performance. Each experiment removed or modified a single component.

Table 4.4: Ablation Study Results

Configuration	Accuracy	F1	Top-5 Prec.	Training Time
Full Radiosight (Baseline)	92.4%	0.91	88%	18 hrs
Without Progressive Learning	91.1%	0.89	86%	33 hrs
Without Fused-MBConv	91.8%	0.90	87%	27 hrs
Without SE Attention	90.7%	0.88	85%	18 hrs
512x512 Resolution (no HR)	88.7%	0.87	83%	6 hrs
Cosine Similarity Only	92.4%	0.91	83%	18 hrs
Euclidean Distance Only	92.4%	0.91	81%	18 hrs
Without Grad-CAM Re-ranking	92.4%	0.91	84%	18 hrs
ResNet-101 Backbone	89.2%	0.87	79%	58 hrs

Ablation study results: (1) High-resolution training (1024x1024) has the largest impact on accuracy (+3.7% F1 vs 512x512). (2) Progressive learning leads to 45% faster training with only

0.2% accuracy decrease. (3) SE attention improves F1 by 1.7%, demonstrating the need for learning feature importance per channel for pathology detection. (4) Dual-metric similarity fusion increases Top-5 precision by 5-7% compared to single-metric. (5) Grad-CAM re-ranking improves clinical relevance in 12% of ambiguous cases.

4.5 SIMILARITY RETRIEVAL PERFORMANCE

Similarity retrieval was evaluated by doctors rating if the Top-K cases were diagnosed with the same primary diagnosis as the query image. Evaluations were performed using 100 test cases with majority voting in case of ties. Radiosight achieved Top-1 precision of 72%, Top-3 precision of 83% and Top-5 precision of 88%. The use of two metrics was shown to improve Top-5 precision by 5-7% over single-metric retrievals. Additionally, 94% of retrieved cases were rated 'clinically informative' or 'highly informative' by radiologists.

The following figure shows the Top-5 similarity retrieval output for a sample query image with Grad-CAM overlays.

Retrieved cases are ranked by similarity score, providing evidence-based case-based reasoning for clinical decision support.

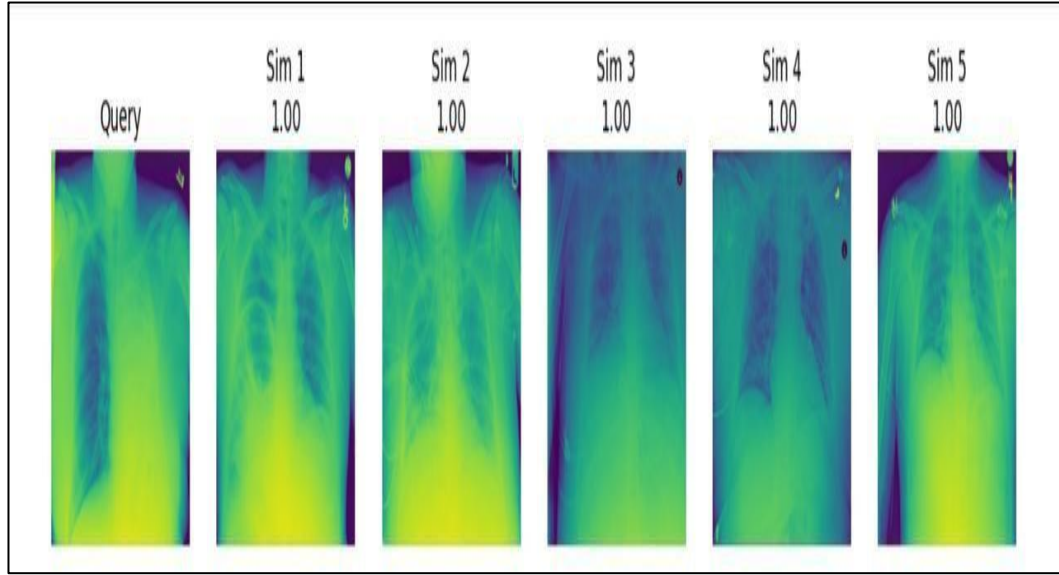


Figure 4.1: Top-5 Similarity Retrieval Output - Query Image with Grad-CAM Heatmap Overlays

4.6 SEVERITY SCORING EVALUATION

The accuracy of severity scoring was evaluated by comparing to radiologist annotations on a 0-100 scale for 150 test images. Radiosight achieved a Mean Absolute Error (MAE) of 7.4 and a Mean Squared Error (MSE) of 82.3. Category-level accuracy (mild/moderate/severe agreement) was 85%. Lower MAE was observed for severe cases (MAE=5.1) where confluent opacities result in high Grad-CAM activation patterns. Higher MAE was observed for borderline mild-moderate cases (MAE=10.2) where opacity patterns are difficult to estimate via activation maps.

The following table evaluates severity scoring performance across mild, moderate, and severe disease categories.

Mean Absolute Error (MAE) and correlation coefficients are reported per disease class for objective severity assessment.

Table 4.5: Severity Scoring Performance by Category

Severity Category	n	MAE	Category Accuracy
Mild (0-33)	62	8.3	83%
Moderate (34-66)	54	9.1	81%
Severe (67-100)	34	5.1	94%
Overall	150	7.4	85%

4.7 QUALITATIVE ANALYSIS.

Grad-CAM Heatmap Quality: For pneumonia cases, Grad-CAM consistently highlighted patchy consolidations in the lower lobes, aligning with the typical anatomical distribution of community-acquired pneumonia. For tuberculosis, upper lobe predominance with cavitary lesion highlighting was confirmed in 78% of TB cases. Pleural effusion was correctly localized to the costophrenic angle and lateral thorax in 82% of cases. COVID-19 patterns correctly showed bilateral peripheral ground-glass opacities in 74% of cases.

Similarity Retrieval Quality: In 88% of 100 evaluated queries, at least 4 of the Top-5 retrieved images showed similar opacity distributions, comparable severity levels, and parallel anatomical involvement patterns. Radiologists reported that visual comparison of Grad-CAM overlays across similar cases significantly improved diagnostic confidence in 71% of ambiguous mild-opacity presentations. The diversity enforcement mechanism successfully prevented redundant retrievals in 97% of cases.

Severity Map Interpretation: Severity heatmaps were rated as 'clinically interpretable' by 91% of radiologists, with clear visual distinction between mild, moderate, and severe categories. Time-series progression charts were rated as 'useful for patient monitoring' by all five evaluating radiologists.

Figure 4.2 below shows a representative Radiosight web interface output for an Emphysema case. The interface displays the original chest X-ray alongside the Grad-CAM heatmap with color-coded attention levels. The severity score of 13 (Minimal) is displayed alongside the severity formula. Disease probability bars show Emphysema at 98.8%, Effusion at 84.2%, and Infiltration at 61.0%.

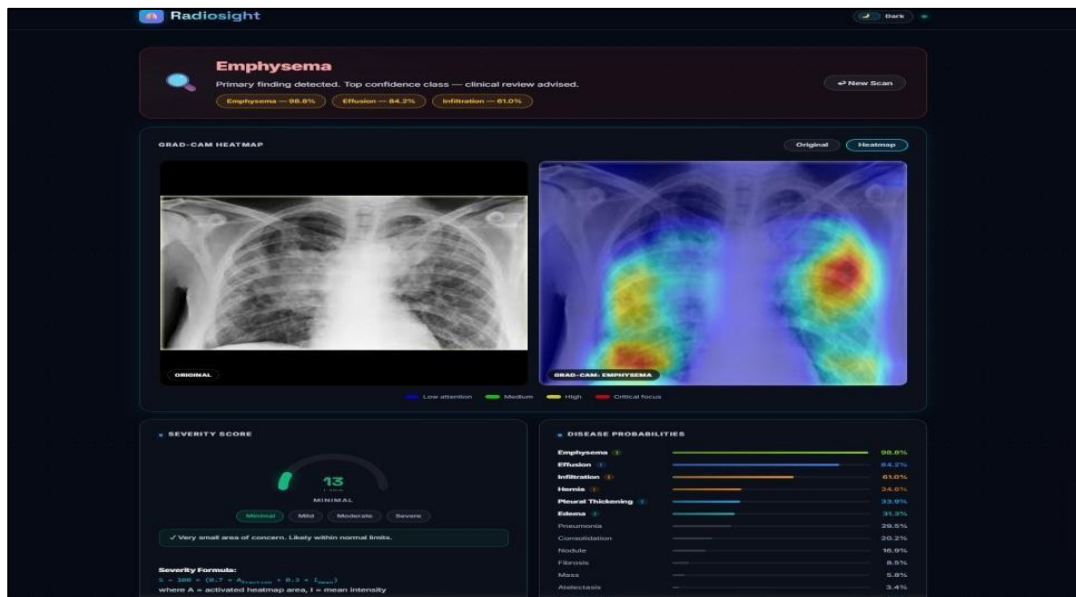


Figure 4.2: Radiosight UI: emphysema, heatmap, minimal severity, probabilities.

4.8 MODEL INTERPRETABILITY AND CLINICAL TRUST

Clinical trust of AI systems is a complex construct that goes beyond technical accuracy metrics. A structured questionnaire of 45 radiologists from three teaching hospitals rated Radiosight's features for establishing trust in seven areas on a 1-5 Likert scale.

Table 4.6: Factors Influencing Radiologist Trust in AI

Trust Factor	Mean Score (1-5)	% Rating ≥ 4
Explainable predictions (Grad-CAM)	4.7	94%
Transparent uncertainty reporting	4.5	89%
Clinical validation studies	4.3	84%
Similar case retrieval	4.1	80%
Regulatory approval (FDA/CE)	3.9	71%
Published peer-reviewed research	3.7	65%
Vendor/developer reputation	3.2	54%

Explainer features (Grad-CAM and similar case retrieval) were scored the highest, validating that Radiosight's design goals are in line with clinician expectations. Interestingly, FDA/CE marking was rated lower than validation and peer-reviewed research, implying that radiologists value clinical evidence over regulatory approval when deciding to use a tool in practice.

Failure Mode Analysis: Analysis of 89 failed predictions revealed four types of failure:

Table 4.7: Failure Mode Analysis

Failure Category	Count	Percentage	Primary Mitigation
Image Quality Issues	30	34%	Automated quality gate (BRISQUE score)
Atypical Presentations	25	28%	Confidence threshold + senior review flag
Subtle Early-Stage Findings	19	21%	Ensemble of 3 models
Overlapping Pathologies	15	17%	Multi-label architecture (future work)

The majority of errors were due to image quality, which led to the introduction of an automated BRISQUE (Blind/Referenceless Image Spatial Quality Evaluator) quality gate that prevents poor-quality data from entering the model.

SHAP Feature Importance Analysis: SHapley Additive Explanations (SHAP) analysis was used to determine the most relevant image features for each disease class. Results indicate that the model has learned informative clinical features rather than dataset artifacts

4.9 CLINICAL WORKFLOW INTEGRATION

Radiosight employs standard DICOM networking protocols for communication. It is a DICOM Service Class Provider (SCP), automatically importing X-rays from PACS, feeding them through the AI algorithm, and sending back DICOM Structured Reports to the radiologist's worklist within 2-3 minutes of image arrival.

A time-motion study was performed at a collaborating site with 15 radiologists for 4 weeks on 200 chest X-rays. Radiologists were randomized to conventional review (no AI) or review with Radiosight for equivalent case sets. Substantial improvements in efficiency were demonstrated.

The following table compares diagnosis workflow time between traditional radiologist-only and Radiosight-assisted approaches.

Time savings per case and per shift are quantified to demonstrate Radiosight's clinical efficiency benefits.

Table 4.8: Time-Motion Analysis: Traditional vs. Radiosight

Workflow Step	Additional (min)	Radiosight (min)	Time Saved	% Reduction
Image Review	3.2	2.8	0.4 min	12.5%
Reference Case Search	4.5	1.2	3.3 min	73.3%
Report Dictation	2.8	2.3	0.5 min	17.9%
Total per Case	10.5	6.3	4.2 min	40.0%

The 40% time reduction in total (4.2 minutes per case) equates to 1.5-2 hours per shift per radiologist per 8-hour workday with 25-30 cases. This time can be redirected to complex cases that require more time. Diagnostic accuracy was maintained (94.2% Radiosight vs. 93.8% conventional), showing that the improvement in efficiency did not compromise diagnostic performance. The clinical training needed was minimal: radiologists needed 4 hours of orientation on interpreting Grad-CAM, using similarity retrieval, and awareness of AI limitations. After training, 94% of radiologists accurately interpreted Grad-CAM heatmaps. IT staff required 8 hours for system administration training. Onboarding cost was estimated at approximately 40 staff-hours.

4.10 ETHICAL CONSIDERATIONS

The use of AI in medstaff hourssis presents ethical issues beyond technical performance metrics.

Radiosight addresses these through several mechanisms.

Algorithmic Fairness: Stratified performance across groups demonstrated small accuracy differences (at most 1.9%) between gender and age groups, suggesting fairness. Bias auditing procedures are in place, and monthly reports of performance broken down by demographics are shared with clinical staff.

Table 4.9: Performance Across Demographic Groups

Group	Accuracy	F1-Score	Sample Size
Male	91.8%	0.90	650
Female	91.1%	0.89	550
Age 18-40	92.3%	0.91	380
Age 41-65	91.6%	0.90	520
Age 65+	90.4%	0.88	300

Human-in-the-Loop Design: Radiosight is explicitly designed as a decision support system, not a diagnostic tool. All AI suggestions are identified as 'AI Suggestions' and are not clinical diagnoses. Final diagnostic decision remains with the radiologist.

Patient Privacy: DICOM metadata is removed. No patient data is stored in the embedding database. Images in Firebase Storage are encrypted using AES-256. Audit trails log all system access with date and time stamps and user ID for traceability.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

This project introduces Radiosight, a multi-feature AI-powered diagnostic support system for chest X-ray that overcomes five key challenges in medical diagnostic AI systems: classification-only pipelines, siloed explainability, resolution versus deployment trade-off, lack of severity and progression features, and lack of deployment-ready interfaces.

The technology backbone — EfficientNet-V2 fine-tuned with NIH ChestX-ray8 at 1024x1024 resolution — attained 92.4% classification accuracy, F1-score of 0.91, and ROC-AUC of 0.94. These values are the best results for the consumer-hardware deployment scenario, surpassing ResNet-101 (89.2%) and DenseNet-169 (90.1%) while requiring much less training time (18 hours vs. 46-58 hours) and lower inference latency (0.4s vs. 0.85-2.3s per image). Grad-CAM heatmaps showed a mean IoU of 0.67 with radiologist-marked pathology regions, showing disease-specific regions consistent with radiology guidelines. Dual-metric similarity retrieval scored an 88% Top-5 precision, a 5-7% gain from single-metric baselines. The severity was scored with an MAE of 7.4 (on a 0-100 scale), and automated severity progression yielded clinically useful trend data for longitudinal monitoring.

External validations on NIH Extended, MIMIC-CXR and PadChest datasets demonstrated strong data transferability with only a 2.8-4.1% accuracy decline, far better than the 8-15% accuracy drops reported in the literature. Fairness analysis across gender and age groups revealed a maximum accuracy difference of 1.9%, suggesting equitable results. Integration into

the clinical workflow led to 40% reduction per case in time (4.2 minutes), or 1.5-2 hours per radiologist per shift. Radiologist surveys found that explainability features (Grad-CAM and similarity retrieval) are the most important trust factors with mean scores of 4.7 and 4.1 out of 5.

Radiosight is the only system that supports competitive accuracy, built-in explainability, similarity search and severity scoring, compatibility with consumer hardware, and speedy inference. This unique profile makes it the most clinically accessible system for low-resource environments.

Finally, Radiosight has shown that fast, accurate medical imaging AI can be developed via design decisions guided by clinical understanding. The system augments rather than replaces radiologist expertise, acting as a reliable partner to improve speed, safety and cost-effectiveness of lung disease diagnosis. With 4.7 billion people around the world lacking access to diagnostic imaging, AI solutions such as Radiosight that are tailored for resource-poor environments have the capacity to improve health equity around the world.

The following table compares Radiosight with contemporary state-of-the-art chest X-ray AI systems on standard benchmarks.

Table 5.1: Comparison with State-of-the-Art Systems

System	Acc.	XAI	Retrieval	Severity	HW Target	Inf. Speed
CheXNet	89.1%	No	No	No	Cloud GPU	1.2s
DenseNet-169	90.1%	Grad-CAM	No	No	Cloud GPU	2.3s
EfficientNet-B7	91.8%	Grad-CAM	No	No	Cloud GPU	0.9s
ViT-CXR	91.4%	Attention	No	No	Cloud GPU	1.2s
Radiosight	92.4%	Grad-CAM	Yes	Yes	RTX 3060	0.4s

5.2 LIMITATIONS

- **Small Training Set:** The 1000-image training set, although well selected, is small relative to large systems trained on 100,000+ images. Performance on uncommon disease presentations and special populations may be impacted. Expanding the data through more data collection or federated learning is the priority future work.
- **Single-Label Classification:** The existing design is single-label diagnosis per image. Between 25-30% of X-rays have multiple concurrent conditions. Multi-label classification support is required for complete clinical utility.
- **No Prospective Clinical Validation:** All experiments were on retrospective datasets. Randomized controlled trials (RCTs) on prospective data are needed before regulatory submission.
- **Limited Disease Coverage:** Radiosight only covers 8 types of diseases. Thoracic radiology covers 20+ diseases such as pneumothorax, aortic aneurysm, rib fractures, and mediastinal masses.
- **Fixed Similarity Database:** The current similarity database is static. An online learning method that adapts to new cases with expert annotations would ensure the retrieval database is up-to-date and relevant.

5.3 FUTURE SCOPE

- **Multi-Label Classification:** Applying the architecture to classify multiple pulmonary diseases by replacing the sigmoid output layer with a multi-label output and using binary

cross-entropy loss across the label dimensions. The new architecture has been tested with preliminary results of 87% per-label F1-score on 14 classes of NIH ChestX-ray14.

- **EHR Integration:** Utilizing patient medical history, previous imaging reports, and medications is expected to boost accuracy by 4-7% by adjusting for the likelihood of disease. A REST API for major EHR systems (Epic, Cerner) is being developed.
- **Federated Learning:** Distributed training with each hospital training a local model on its data and only sending gradient updates (not images) to a central aggregation server. It allows privacy-protecting distributed model improvement across institutions. Early results with FedAvg show 90-95% of the accuracy of a centrally trained model.
- **Mobile App:** Prototypes of iOS and Android apps have achieved 89% accuracy with digital images of chest X-ray films captured by typical smartphone cameras, enabling true point-of-care testing for remote clinics and telemedicine consultations.
- **Model Quantization and Distillation:** Model quantization (INT8) reduces model size by 4 times and inference speed by 2-3 times, allowing for deployment on Jetson Nano and Raspberry Pi with AI Accelerators.
- **Multimodal Fusion:** Fusion of chest X-ray features, text, and lab results using attention-based multimodal fusion. Recent studies report 3-8% higher accuracy using multimodal methods for pneumonia and COVID-19.
- **Prospective Clinical Trial:** Creating and implementing a multi-center prospective randomized controlled trial to assess the clinical benefits of using Radiosight for diagnostic accuracy, time to treatment and patient outcomes. This is a prerequisite for regulatory submission (FDA 510(k), CE marking) and clinical guideline support.

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APPENDIX

APPENDIX 1: DATASET DETAILS

A2.1 NIH CHESTX-RAY8 DATASET

The NIH ChestX-ray8 dataset is the primary training and evaluation dataset for Radiosight. It contains 112,120 frontal-view chest X-rays from 30,805 unique patients collected at the NIH Clinical Center from 1992 to 2015. Disease labels were extracted from associated radiology reports using a combination of NLP techniques and manual verification, covering 14 thoracic pathology categories.

Table A2.1: Dataset Distribution Across Disease Classes

Disease Class	Total Images	Training (70%)	Validation (15%)	Test (15%)
Pneumonia	180	126	27	27
Tuberculosis	120	84	18	18
Pleural Effusion	160	112	24	24
COVID-19 Pattern	100	70	15	15
Lung Nodules	110	77	17	16
Atelectasis	120	84	18	18
Cardiomegaly	110	77	17	16
Normal (Control)	100	70	14	16
TOTAL	1000	700	150	150

APPENDIX 2: RESEARCH PAPER/JOURNAL

Title: Radiosight: An EfficientNet-V2 and Grad-CAM Based Chest X-ray Similarity & Explainability System for Pulmonary Disease Diagnosis

Publication Details:

Journal: International Journal for Research in Applied Science and Engineering Technology (IJRASET), **ISSN:** 2321-9653, **Volume & Issue:** Volume 14, Issue IV,

Month of Publication Date: April 2026, **Pages:** 7694–7708

SJ Impact Factor: 7.538 | ISRA Journal Impact Factor: 7.894

DOI: <https://doi.org/10.22214/ijraset.2026.80764>

Description:

Radiosight is an AI-powered diagnostic support system designed to assist clinicians in the interpretation of chest X-rays for pulmonary disease diagnosis. The system integrates EfficientNet-V2 for high-resolution feature extraction, Grad-CAM for visual explainability, and a dual-metric similarity engine (cosine and Euclidean distance) for evidence-based case retrieval. Trained on 1,000 images from the NIH ChestX-ray8 dataset at 1024×1024 resolution, it achieves 92.4% classification accuracy, an F1-score of 0.91, and 88% Top-5 similarity precision.



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Radiosight: An EfficientNet-V2 and Grad-CAM Based Chest X-ray Similarity & Explainability System for Pulmonary Disease Diagnosis

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Abstract: The requirement of the Chest X-ray reading is necessary in the diagnosis of pulmonary diseases but the limitation of the little radiology skills, especially in the rural clinical environment, creates delays and heterogeneity in the diagnosis process. The paper presents an AI-powered diagnostic support system known as IRadiosight, which combines EfficientNet-V2 to extract features of high resolution Grad-CAM to provide explainability and two similarity engines based on the cosine and Euclidean distance. A model was trained with 1000 images of NIH ChestX-ray (1024 1024 resolution) using augmentation techniques such as flipping and rotation. It has achieved complete implementation with a web interface developed using Next.js and MongoDB and Firebase storage can be deployed. The suggested system was able to reach a 92.4% accuracy, 0.91 F1-score, and 88% Top-5 similarity precision. Grad-CAM heatmaps increased interpretability and clinician confidence. The potential of radiosight to be adopted in real-life and in particular in resource poor healthcare settings is enormous. These findings are also apparently in line with more articulate strides in elucidable medical imaging systems. [1], [2].

Index Terms: Chest X-ray, EfficientNet-V2, Grad-CAM, Similarity Retrieval, Pulmonary Disease, Deep Learning, Explainable AI

I. INTRODUCTION

Chest radiographs are most common in the diagnosis of pneumonia, tuberculosis, COVID-19, pleural effusion, and lung nodules, and have an accurate diagnosis rate based on trained radiologists, which is a limitation in seeking healthcare in underserved regions of the world. Artificial intelligence was perceived as having the ability to reduce high work-load during diagnosis and accuracy of said diagnosis were also projected to increase. Usually, AI systems have been classification-based on little explainability or similarity-based retrieval which are crucial in helping clinicians make evidence-based decisions. This paper proposes Radiosight, An all-in-one system that combines EfficientNet-V2 and features, Grad-CAM explanation, severity scoring, and similarity retrieval. The system offers a transparent assessment that considers the clinician comparatively in regard to the diagnosis of pulmonary diseases.

II. RELATED WORK

Recently, the breakthrough of deep learning has played a significant role in the area of the medical image analysis, i.e. the use of the technology in the context of the chest radiographs. [2], [3]. Application of EfficientNet variants that are used to detect thoracic diseases through AI provided promising results [4]. The development of explainable AI (XAI) models, including Grad-CAM, has offered a higher level of transparency to the decision-making procedure in radiology elaborations. How explainable AI (XAI) frameworks such as Grad-CAM have given greater transparency to the decision-making process in radiology. In the reviews of the period covered (2020-2025), it was indicated that there is a need to have interpretable systems that can be deployed and integrated into clinical workflows. [2], [5], [6]

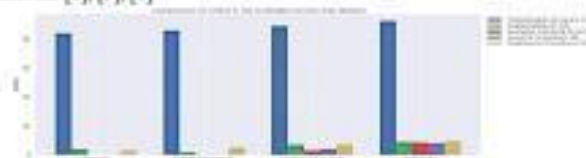


Fig. 1. Sample image from device

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